Stubbornness and Social Power in Opinion Dynamics: the French–Harary–DeGroot and Friedkin–Johnsen Models

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Abstract

A stubborn agent, which never changes its opinion, is the extreme case of two different mechanisms of resistance to social influence: a self-weight in the French–Harary–DeGroot (FDG) model and an attachment to a fixed prejudice in the Friedkin–Johnsen (FJ) model. Following the tutorial by Proskurnikov and Tempo, this report compares the two models on the same small networks and studies how these mechanisms affect convergence, the final opinions and the influence of each agent, measured by the social power and by the influence centrality. A self-weight acts as inertia: it ~~slows an agent down, makes the network aperiodic and earns influence gradually~~ties an agent to its current opinion, which still follows the others, and on a strongly connected network it does not prevent a consensus. A prejudice acts as an anchor: ~~a single prejudiced agentthat influences everybody makes~~ it ties an agent to a fixed opinion; when every agent is influenced by a prejudiced agent, the FJ model ~~asymptotically stable and takes the whole influence centrality, while, when every agent is prejudiced , different prejudices keep the group divided on any network~~is asymptotically stable on any network, and the group agrees only if all the prejudiced agents share the same prejudice. Writing the FJ model as an FDG model on an augmented graph, with the prejudices as virtual stubborn agents, shows that the two models differ in what is a state: the opinion of a stubborn agent is ~~a memory inside~~ part of the state in the FDG model and an external input in the FJ model.

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